Problem 25.1

Calculate the present value of a continuous annuity of 1,000 per annum for 8 years at:

(a) An annual effective interest rate of 4%;

(b) A constant force of interest of 4%.

Solution

For a continuous annuity paying at a rate of 1,000 per year for 8 years, the present value is

$$1000\bar{a}_{\overline{8}|} = 1000\left(\frac{1 - e^{-\delta 8}}{\delta}\right),$$

where δ is the force of interest.

Part (a)

The annual effective interest rate is 4%, so

$$i = 0.04.$$

The corresponding force of interest is

$$\delta = \ln(1 + i) = \ln(1.04).$$

Also,

$$e^{-\delta 8} = (1.04)^{-8}.$$

Thus,

$$PV = 1000\left(\frac{1 - (1.04)^{-8}}{\ln(1.04)}\right).$$

Therefore,

$$PV = 6866.5195.$$

So, for part (a),

$$\boxed{6866.52}.$$

Part (b)

The constant force of interest is 4%, so

$$\delta = 0.04.$$

Then

$$PV = 1000\left(\frac{1 - e^{-0.04(8)}}{0.04}\right).$$

Thus,

$$PV = 1000\left(\frac{1 - e^{-0.32}}{0.04}\right).$$

Therefore,

$$PV = 6846.274073.$$

So, for part (b),

$$\boxed{6846.27}.$$